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A vision-based method for automatic tracking of construction machines at nighttime based on deep learning illumination enhancement

Bo Xiao [a], Qiang Lin [b], Yuan Chen [c,*]

[a] Department of Civil and Environmental Engineering, University of Alberta, Edmonton T6G 2R3, Canada
[b] School of Civil Engineering, Chongqing University, Chongqing 400045, PR China
[c] College of Management and Economics, Tianjin University, Tianjin 300072, PR China

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ABSTRACT

Nighttime construction has been widely conducted in many construction scenarios, but it is also much riskier due to low lighting conditions and fatiguing environments. Therefore, this study proposes a vision-based method specifically for automatic tracking of construction machines at nighttime by integrating the deep learning illumination enhancement. Five main modules are involved in the proposed method, including illumination enhancement, machine detection, Kalman filter tracking, machine association, and linear assignment. Then, a testing experiment based on nine nighttime videos is conducted to evaluate the tracking performance using this approach. The results show that the method developed in this study achieved 95.1% in MOTA and 75.9% in MTOP. Compared with the baseline method SORT, the proposed method has improved the tracking robustness of 21.7% in nighttime construction scenarios. The proposed methodology can also be used to help accomplish automated surveillance tasks in nighttime construction to improve the productivity and safety performance.

1. Introduction

Construction at nighttime is a common practice in many construction scenarios including pavement maintenance, highway construction, and railway construction [1]. Compared with daytime construction, nighttime construction has some obvious advantages [2]. For instance, operating outdoor construction activities (e.g., for the pavement project) at nighttime can reduce the interferences with traveling vehicles to avoid traffic congestions. Furthermore, the working environment of nighttime construction is less complex than that during the daytime (e.g., fewer pedestrians and traffics), which can improve the crew productivity to some extent. Especially, in the summer months, the temperature at nighttime is much lower than that at daytime. The cooler construction condition could speed up material delivery cycles and shorten machinery idling time.

In addition, awareness of construction safety at nighttime has been a major concern in the industry since nighttime construction naturally creates hazardous working conditions. The results from the study of Arditi et al. [3] indicated that the safety risks at nighttime can be five times higher than daytime construction due to some significant factors: a) the lower illumination conditions; and b) workers and machine operators are prone to be fatigue. By analyzing 20,997 construction industry accident records in the United States from 1997 to 2012, Kang et al. [4] identified 22.8% of construction accidents belonged to struck-by accidents, which is the second most influential factor for construction safety. Therefore, monitoring construction machines at nighttime is beneficial to avoid struck-by hazards between works and machines, which is important for facilitating a relatively safe working environment. Currently, the manual method still acts as the primary approach for monitoring nighttime construction, but its usage is error-prone and time-consuming. As such, automatically tracking construction machines with retrieving the trajectory information can be treated as an alternative way to help identify the potential collisions and reduce the risks of struck-by accidents in nighttime construction [5].

Recently, tracking construction machines by using vision-based methods has become a promising way considering cameras are less costly and simply to install [6]. Extensive research studies have been conducted on tracking construction machines from videos for evaluating crew productivity and enhancing site safety in the area of construction engineering and management. For instance, Chen et al. [7] automatically calculated the productivity of excavators in earthmoving works by tracking excavators using vision-based methods. Kim et al. [8] proposed an automatic method for analyzing the productivity of tunnel earthmoving by means of tracking excavators and dump trucks. Yan et al. [9]

* Corresponding author.
E-mail addresses: bxiao2@ualberta.ca (B. Xiao), linqiang@cqu.edu.cn (Q. Lin), yuanchen_2020@tju.edu.cn (Y. Chen).

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adopted a struck-by accident monitoring method based on tracking heavy machines and workers from videos in order to enhance site safety.

It can be seen that most of the existing vision-based methods for tracking construction machines tend to focus on daytime construction [10], but it is difficult to achieve reliable performance when using these approaches in nighttime scenarios. Tracking construction machines from nighttime videos is challenging due to the limited visibility and significant illumination variations [11]. In nighttime construction, the lighting conditions are usually poor and construction machines are prone to be partially invisible. Meanwhile, the illumination variations can easily create blurs of construction objects, which will trigger some difficulties in distinguishing one construction machine from others through tracking. Therefore, it is urgent and necessary to develop a more robust appearance model for automatically tracking construction machines in nighttime scenarios.

The main objective of this research is to propose a vision-based method for automatic tracking of construction machines at nighttime. Specifically, this approach can enhance the image quality for tracking in low lighting conditions and explore the robust appearance model for describing machine objects when facing illumination variations. The proposed tracking method is expected to produce stable and precise bounding boxes of construction machines from nighttime videos in order to generate trajectory information. By integrating the method developed in this study, many surveillance tasks in nighttime construction can be accomplished automatedly including crew productivity evaluation, machine idling analysis, and collision alerts, which will make nighttime construction more efficient and safer.

2. Literature review

Automatic tracking of construction entities by vision-based methods is an important topic in the research community, which is useful for project monitoring, materials transportation, and site safety. In this section, the literature regarding the vision-based tracking methods and their applications in construction will be reviewed. Since this research tries to integrate illumination enhancement methods for nighttime tracking, the development of the image illumination enhancement will also be reviewed comprehensively.

2.1. Vision-based tracking methods

Vision-based tracking refers to track the interested objects from videos to acquire the trajectory information for surveillance, which is a fundamental problem in the computer vision field [12]. Given the initial pixel position of the objectat the first frame, the vision-based tracker will start to predict the pixel positions of this object at the subsequent frames [13]. In general, a typical vision-based tracking method consists of three main components including the motion model, the appearance model, and the observation model [14].

In the tracking process, the motion model generates a set of candidate windows at the current frame based on the tracking window in the previous frames. Dense sampling is a widely adopted motion model, by means of which all possible candidacy window positions are enumerated through a sliding window mechanism [15]. Another common motion model is particle filtering, which estimates the potential candidate windows by the particle distribution [16]. The appearance model in vision-based tracking aims to extract features from the candidate windows. The classic appearance models include the histograms of oriented gradients [17], Haar-like features [18], and the covariance region descriptor [19], etc. The convolutional neural network (CNN) [20] has also been adopted as the feature extractor for tracking common life objects (e.g., persons, cars, and animals). Furthermore, the observation model judges all candidate windows based on their features to produce the tracking result in the current frame as well as update the observation model. The observation model can be the similarity between features [21] or the machine learning classifiers (e.g., the support vector machine [22]).

Recently, object detection has been integrated into the task of object tracking and this kind of method is called tracking-by-detection [23]. The tracking-by-detection comprises three main steps including: 1) detecting objects at each frame by the backbone detector, 2) converting the detected objects to features by the appearance model, and 3) associating the object features across frames to produce tracking results. For example, Rezatofighi et al. [24] introduced a vision-based tracking method based on the joint probability data association. Bewley et al. [25] developed a simple online sort tracking method based on Kalman filter tracking and pixel location similarity. Bochinski et al. [26] proposed a high-speed tracking-by-detection method without using image information, which has achieved the superfast tracking speed. Most existing vision-based tracking methods aim to track common life objects in general scenarios.

2.2. Vision-based tracking methods in construction

In the area of construction engineering and management, many research studies have focused on the development of vision-based methods specifically for construction scenarios. Xiao and Kang [27] have proposed a vision-based method of tracking construction machines by introducing image hashing as the appearance model. Angah and Chen [28] have developed a deep learning-based construction worker tracking method by integrating Mask R-CNN [29] and trajectory rematching. Roberts et al. [30] have proposed a vision-based tracking method of construction workers to estimate worker body poses, which has achieved an accuracy of 72.6%. Kanstantinou et al. [31] integrated three tracking models (i.e., the adaptive model, the appearance model, and the prediction model) for tracking multiple workers from a complex environment. Zhu et al. [32] employed the latent support vector machine and particle filtering to track construction machines and workers. However, these developed vision-based tracking methods are mainly focusing on tracking construction objects in daytime construction and cannot be directly applied to automatic tracking in nighttime construction due to the low visibility and huge illumination changes in nighttime scenarios.

The application of vision-based tracking methods can be broadly divided into two primary aspects: productivity estimation and safety monitoring [33]. For productivity estimation, these methods can be used to retrieve the movement information of construction machines in cyclic operations (e.g., earthmoving works). Golparvar-Fard et al. [34] proposed a framework that integrated vision-based tracking and action recognition to measure the earthmoving productivity. Rezazadeh et al. [35] combined vision-based tracking with detection for automatically calculating the dirt-loading cycles. Roberts and Golparvar-Fard [36] adopted an end-to-end system of combining vision-based detection, tracking, and activity recognition for monitoring the machine productivity in earthmoving projects.

To facilitate construction safety, vision-based tracking methods can be used to produce the trajectories of machines and workers to avoid potential collisions on the site. Son et al. [37] proposed a real-time vision-based warning system to prevent collisions between workers and heavy machines by tracking the two types of objects. Kim et al. [38] developed a vision-based site safety assessment system to monitor the struck-by accidents by tracking equipment and workers simultaneously. To be noted, the abovementioned construction applications are currently proposed for daytime construction scenarios because few studies have developed a reliable vision-based method that is technically designed for tracking construction machines at nighttime.

2.3. Image illumination enhancement

Enhancement of low lighting images by automatic methods has attracted much attention from the computer vision community. Compared with manually adjusting the image contrast or brightness, automatic methods are more convenient to be applied in dealing with a large number of images. Existing approaches for image illumination enhancement can be generally categorized into the retinex-based methods and learning-based methods [39]. In the retinex theory [40], an image can be treated into the pixel-wise produce of reflectance and illumination, and the reflectance component is a plausible approximation of the enhanced image. Therefore, the illumination enhancement problem is formulated as an illumination estimation problem in the retinex-based methods. Cai et al. [41] used a joint intrinsic-extrinsic prior model for estimating the illumination of an image. Fu et al. [42] employed a weighted variational model to estimate the reflectance and illumination of an image in order to improve the image lighting conditions. Wang et al. [43] developed a naturalness preserved enhancement algorithm for improving non-uniform illumination images. However, the retinex-based methods have less capability to enhance color images since color is easily distorted locally, which will lead to some image details lost in the enhancement process.

The learning-based methods treat the image illumination as a distribution estimation problem, which means the high lighting image can be estimated by adding a certain distribution to the original low lighting image [44]. This kind of methods are data-driven so that they can learn the illumination distribution from pairs of low lighting and highlighting images by deep neural networks. Lore et al. [45] proposed an encoder-decoder network named LLNet to enhance the natural low-light images. Chen et al. [46] built an image illumination enhancer based on the framework of two-way generative adversarial networks. Gharbi et al. [47] developed a novel neural network architecture based on bilateral learning and achieved the real-time processing speed. It can be seen that the learning-based methods can keep most of image details in the procedure of image illumination enhancement. Currently, the image illumination enhancement methods are mainly adopted for the use of the social medias and have not been investigated in tracking construction entities in nighttime scenarios.

2.4. Research challenges and objectives

As reviewed in the previous subsections, tracking construction machines at nighttime by vision-based methods is the fundamental step for automatic applications in nighttime construction. A robust vision-based tracking method for construction machines can be adopted in many surveillance tasks in order to make nighttime construction safer and more efficient. However, existing methods of tracking construction machines at nighttime have two major technical challenges. Firstly, the backbone detector adopted in exiting tracking methods cannot produce precise detection results from nighttime videos because most image data used for training the backbone detector are collected at daytime. Secondly, the appearance models adopted in existing methods are difficult to extract distinguished features from construction objects in nighttime videos. This is because insufficient lighting conditions in nighttime videos easily triggers the object invisibility, motion blur, and illumination variations issues.

The overall objective of this research is to develop a vision-based method for automatically tracking construction machines at nighttime. To overcome aforementioned research challenges, the proposed method employs the deep learning illumination enhancement to preprocess the input videos in advance. By doing that, some construction machine objects that are partial invisible in the original images will become identifiable and detectable. Meanwhile, the CNN feature extractor is adopted in the proposed method as the appearance model, which helps to distinguish machine objects in nighttime videos when facing the challenge of illumination variations and motion blurs.

3. Methodology

In this section, the overall framework of the proposed methodology is introduced firstly. Then, five main modules involved in this vision-based method are described in detail, that is, illumination enhancement, machine detection, Kalman filter tracking, machine association, and linear assignment.

3.1. Overall framework

The overall framework of the proposed methodology is depicted in Fig. 1. First, the frame sequences derived from nighttime videos are inputted into the illumination enhancement module, which is developed based on an encoder-decoder deep neural network to repair the low lighting images. Then, the deep learning detection is executed on the enhanced frames to identify construction machines with the pixel position and categorical information in the machine detection module. The detection windows in the previous frame are used to initialize the Kalman filter trackers in the third module, which can produce the tracking windows for the current frame. Next, the machine association module can relate detection windows to tracking windows at the current frame based on location similarity and CNN feature similarity to construct the association matrix. Subsequently, the tracking problem is converted to a linear assignment problem, which can be solved by the Hungarian algorithm to produce the final tracking results. In summary, the technical novelty of the proposed methodology consists of two primary parts: a) integrating the deep learning illumination enhancement to repair the nighttime videos, which can overcome the low visibility issue in construction machine tracking at nighttime. b) employing a CNN feature extractor in the machine association module as the appearance model for representing construction machines.

3.2. Illumination enhancement

The purpose of this module is to address the problem of low lighting conditions in nighttime construction videos. Insufficient illumination can damage the visual quality and degrade the performance of object detection algorithms for identifying construction machines from videos. However, directly increasing the contrast and brightness of nighttime frames can lead to the overexposure issue, and details will be lost in the shadow and darkest area of the frame. Therefore, an algorithm for the deep learning illumination enhancement is employed to repair the nighttime frames and further produce the illumination enhanced frames with keeping most of details.

Specifically, the global illumination-aware and detail-preserving network (GLADNet [48]) is selected in this research due to its state-of-the-art performance on public datasets including LIME [49], DICM [50], and MEF [51]. The architecture of the GLADNet, as depicted in Fig. 2, comprises two adjacent steps, that is, illumination distribution estimation and detail reconstruction. As for the former, the input images are resized to 96×96 by the nearest-neighbor interpolation. Then, feature maps are passed through an encoder-decoder network to estimate the global illumination of the image. The encoder networks use the CNN to down-sampling features, while the decoder networks use the resized-CNN to up-sampling feature. The detail reconstruction step adopts the concatenation to combine the feature maps of the outputs derived from the global illumination step with the input image. Then, the concatenated feature maps are processed by three convolutional layers with ReLUs in order to preserve more details of the input image.

In the proposed method, all frames of the nighttime video are processed with the GLADNet to obtain the enhanced frames, which will be regarded as the inputs of the machine detection module. Furthermore, Fig. 3 illustrates the original nighttime frame, the frames enhanced by GLADNet and by adjusting brightness, respectively, in order to clearly display differences in the performance of illumination enhancement by various approaches. Compared with directly adjusting the image brightness, the use of GLADNet is able to greatly improve lighting conditions of this image while keeping most of details.

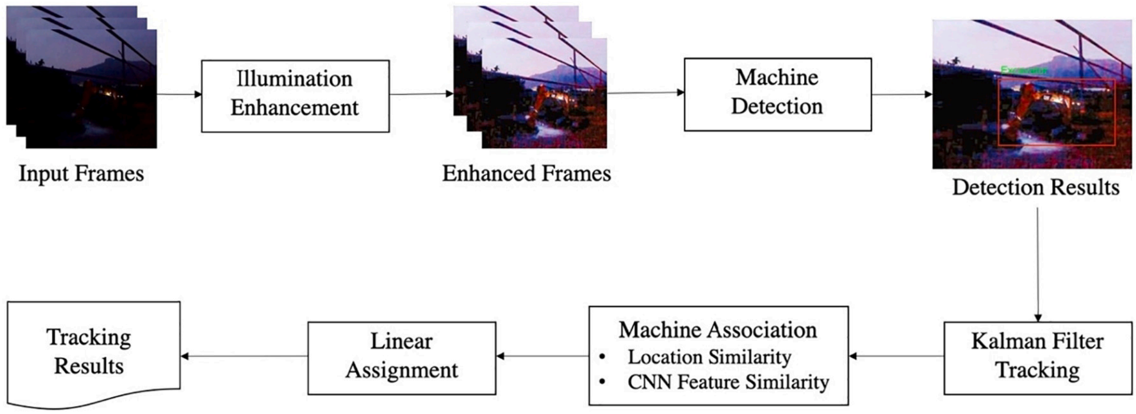


Fig. 1. The overall framework of the proposed methodology.

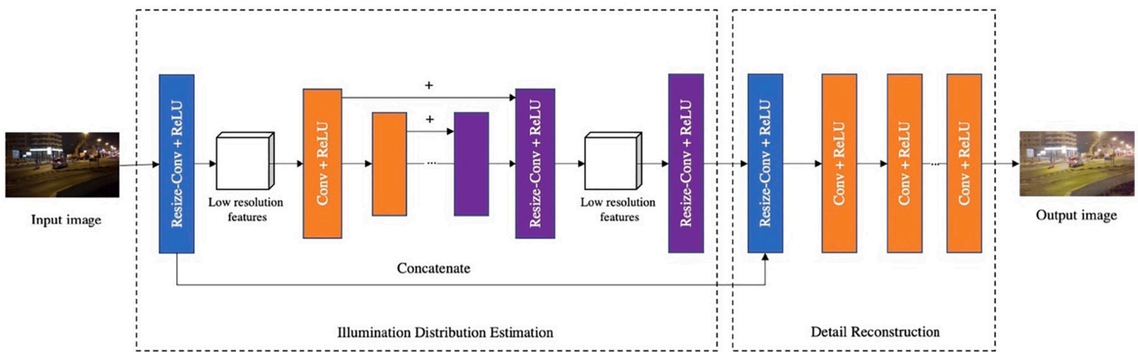
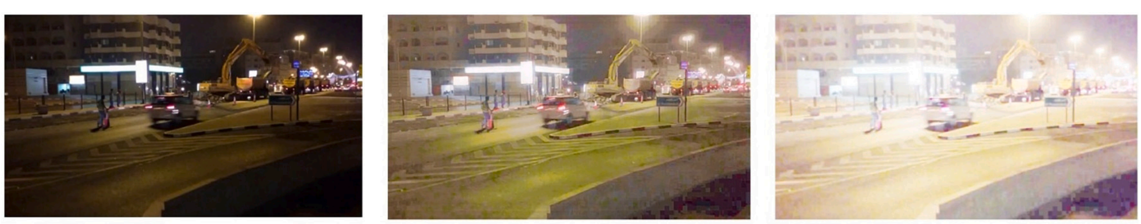


Fig. 2. The architecture of GLADNet.



(a) Original nighttime image (b) Image enhanced by GLADNet (c) Image enhanced by adjusting brightness

Fig. 3. Performance comparison of image enhancement.

3.3. Machine detection

In this module, the enhanced frames are processed with the algorithm for deep learning object detection to retrieve the pixel location and categorical information of pre-defined categories of construction machines. The robust detector can produce stable and precise detection windows, which will facilitate a good tracking performance. In the present research, the deep learning algorithm YOLO-v3 [52] is selected as the backbone because it has achieved the mean average precision (mAP) of 57.9% on the COCO benchmark [53] with more than real-time detection speed (30 frames per second on a GTX Titan X GPU). The architecture of the YOLO-v3 detector is illustrated in Fig. 4. YOLO-v3 is a fully convolutional detection network that contains 53 convolutional layers where each followed by batch normalization layer and leaky ReLU activation. Also, the YOLO-v3 algorithm adopts multiple-scale CNN architecture specifically for detecting small objects, which is much helpful in construction scenarios considering it is common to find this kind of objects in construction videos.

When using the YOLO-v3 algorithm, an annotated dataset is necessary for training deep learning object detectors. This study will select the Alberta Construction Image Dataset (ACID [55]), which is an image dataset of standard construction machines for the task of object detection. The ACID contains 10,000 annotated images of construction machines that can be divided into ten categories, including excavator, compactor, dozer, grader, dump truck, concrete mixer truck, wheel loader, backhoe loader, tower crane, and mobile crane. The images of ACID are collected from the different construction scenarios to ensure the high degree of diversity and further avoid the overfitting problem of deep learning object detectors. By training with data from the ACID, the proposed tracking method can simultaneously detect construction machines that belong to the abovementioned categories from the enhanced images.

3.4. Kalman filter tracking

When a new machine object is detected in a frame, a Kalman filter tracker will be initialized to track only this machine and a unique identification (ID) is assigned to the tracklet. In the next frame, these

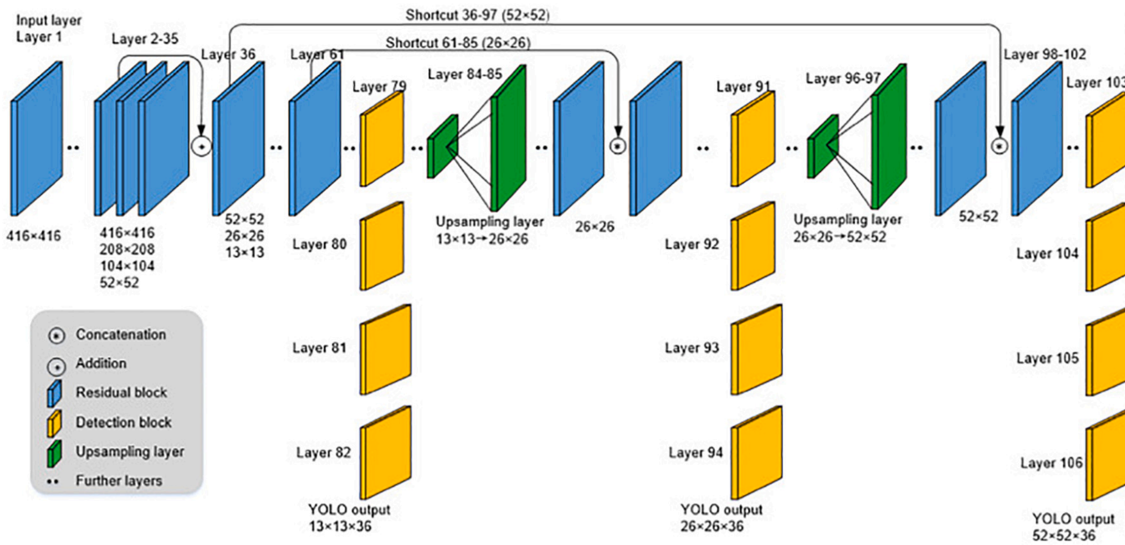


Fig. 4. The architecture of the YOLO-v3 detector (image reprint from [54] with permission).

trackers belonging to all the detected machines will produce the tracking results to associate the detection results at the frame. The Kalman filtering is an algorithm that utilizes continuous measurements over time, and produces estimation of the current statement [56]. Fig. 5 shows the overall process of Kalman filter tracker, where $X_{k|k-1}$ refers to the estimate of the tracking state at time stamp k until the $k-1$th measurement, $P_{k|k-1}$ is the corresponding uncertainty, and y_k is the measurements at the time stamp k. Generally, the Kalman filter tracker works in two steps: a) The prediction process produces the estimation of the current statement $X_{k|k-1}$ based on previous predictions and measurements, although the current statement is uncertain; and b) Once the measurement of the current statement y_k is provided, the Kalman filter can be updated by the state transition model to output the final tracking result $X_{k|k}$.

Similar to the previous study [25], the state of each object is modeled as:

$$X = [x_c, y_c, w, h, u, v] \tag{1}$$

where, x_c and y_c mean the horizontal and vertical coordinates of the central point of the object; w is the width and h is the height of the object; u and v represent the velocity of the object on the horizontal and vertical axis, respectively. When a detection window is associated with a tracking window, the detection box will be used to update the Kalman filter tracker. If no detection windows are associated to this object, the tracker will simply update its state by using the linear velocity model.

3.5. Machine association

In each frame, the YOLO-v3 algorithm will produce the detection windows and Kalman filter tracking will predict tracking windows. The main purpose of the machine association module is to construct matrixes to represent the similarity between detection windows and tracking

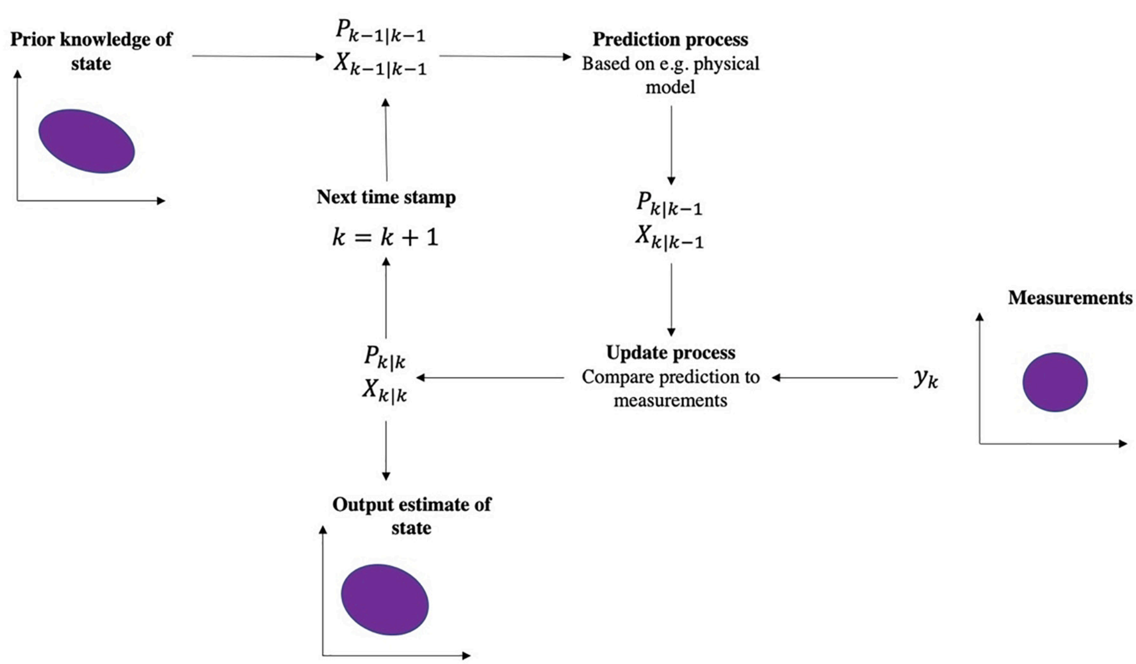


Fig. 5. The overall process of Kalman filter tracker.

windows. Afterwards, the detection window can be assigned with a tracking ID number once it is associated with a tracking window in the linear assignment module.

To achieve this goal, the position similarity and CNN feature similarity are combined to describe the relationship between the detection window *i* and the tracking window *j*. The position similarity is calculated by the intersection over union (IoU), which is defined in Eq. (2).

$$IoU(i,j) = \frac{Area(i) \cap Area(j)}{Area(i) \cup Area(j)} \tag{2}$$

The CNN feature similarity evaluates the visual similarity between two object windows, which is an efficient appearance model to distinguish machine objects when facing illumination variations and motion blurs in nighttime scenarios. First, all detection and tracking windows are resized to the size of 224×224 and inputted to a ResNet50 neural network [57] (pretrained on ImageNet). Then, a feature vector with the size of 500 by 1 can be extracted from the fully connected layer of the ResNet50 to represent each input object window. The CNN feature similarity *k* between the detection window *i* and the tracking window *j* can be calculated as the cosine similarity of their corresponding feature vectors (see Eq. (3)).

$$k(i,j) = \frac{v(i)v(j)'}{\|v(i)\|\|v(j)\|} \tag{3}$$

where, $v(i)$ and $v(j)$ indicate the feature vector processed by ResNet50 of the detection window *i* and the tracking window *j*, respectively. And the $\|v(i)\|$ is the norm of the vector $v(i)$.

Based on this, the similarity between the detection window *i* and the tracking window *j* can be calculated as the linear combination of position similarity and CNN feature similarity as Eq. (4).

$$similarity\ (i,j) = \alpha \times IoU(i,j) + (1-\alpha) \times k(i,j) \tag{4}$$

where, α is a constant and its value is equal to 0.5 in this study.

As such, an association matrix *A* can be built by integrating the similarity between each pair of detection window and tracking window. The size of the association matrix is the number of detection windows by the number of tracking windows.

3.6. Linear assignment

The linear assignment module aims to assign a tracking ID to each detection window by associating it with a tracking window. The tracking problem is then converted to a linear assignment problem and can be formulated as follows:

$$maximize: \sum_{i=1}^{a} \sum_{j=1}^{b} A[i,j] \times x_{ij} \tag{5}$$

$$A[i,j] = similarity(detection\ window\ i, tracking\ window\ j) \tag{6}$$

$$x_{ij} = \begin{cases} 1, if\ the\ detection\ window\ i\ is\ assigned\ to\ the\ tracking\ window\ j \\ 0, if\ the\ detection\ window\ i\ is\ not\ assigned\ to\ the\ tracking\ window\ j \end{cases} \tag{7}$$

which is subject to:

$$\sum_{i=1}^{a} x_{ij} = 1 when\ j = 1, \dots, b \tag{8}$$

$$\sum_{j=1}^{b} x_{ij} = 1 when\ i = 1, \dots, a \tag{9}$$

The variables *a* and *b* are the number of detection and tracking windows at the current frame, respectively. The first constraint (Eq. (8)) means each detection window can only be assigned with one tracking window, while the second constraint (Eq. (9)) indicates each tracking window can only be associated with one detection window. This above linear assignment problem can then be solved by the Hungarian algorithm [58] to assign tracking IDs to detection windows.

In the linear assignment module, if a detection window *i* is successfully matched with a tracking window *j* by the Hungarian algorithm and the *similarity* (*i*,*j*) is larger than 0.5, the tracking window *j* will be regarded as the tracking result and the detection window *i* will be used to update the corresponding Kalman filter tracker. If the detection window *i* cannot match with a tracking window *j* or the *similarity* (*i*,*j*) is not larger than 0.5, a new tracking ID will be assigned to the detection window *i* and a new Kalman filter tracker will be initialized in this module. Also, the detection window *i* will be the tracking output. If a Kalman filter tracker cannot associate any detection window in ten continuously frames, this tracklet and the corresponding tracking ID will be destroyed.

4. Experiments and results

The experiments and results are presented in this section. First, the implementation of the proposed method is introduced. Then, the experimental setup, evaluation metrics, and experimental results are reported in details.

4.1. Implementation of the proposed method

At the implementation stage, the YOLO-v3 algorithm was trained 200,000 iterations on the ACID dataset with the batch size of 32. The proposed method was implemented in Python language and adopted the Opencv library for video I/O. The GLADNet was programmed by the Pytorch library. The YOLO-v3 was originally implemented by C language and integrated in the proposed method by means of the Python wrapper. The Kalman filter tracking was built on the filterpy library and the Hungarian algorithm was implemented by the scikit-learn library. For the hardware configuration, the proposed method was tested on a computer that has one NVIDIA GTX 1080Ti GPU, an Intel Core i9-7920×@2.9 Hz CPU, and two 32 GB memory cards.

4.2. Experimental setup

To evaluate the performance of the proposed method, nine testing videos were collected from the nighttime construction scenarios (e.g., earthmoving and pavement maintenance works), and each video contained one or several construction machine movements. Fig. 6 shows the example images of each testing video and Table 1 summarizes the information of each testing video in terms of the number of frames, video duration, video resolution, construction machines appeared, and the video brightness. The video brightness was assigned by a value between 0 and 255, which showed the lighting conditions of this video. First, all frames of the video were converted into the grayscale images and the average of grayscale pixel values was the brightness of this frame. As such, the video brightness was equal to the average brightness value of all frames, and the lower value indicated the lower lighting conditions in this video. Then, all videos were manually annotated at the frame level as the ground truth for evaluating the proposed method. In each frame, the annotation included the frame number, ID number, and the object pixel coordinates. The annotations are in the MOT16 format [59] and used for a comparison of the tracking results produced by the proposed method in the evaluation process.

In the experiments, the tracking method SORT [25] was selected as the baseline method to compare tracking results of the testing videos with the proposed method in this study because SORT has achieved the state-of-the-art tracking performance on the MOT16 benchmark. The SORT method associates the detection results across frames only based on the position similarity (IoU). Also, it adopts the Kalman filter for tracking and Hungarian algorithm for producing tracking results, which

Fig. 6. Example images of testing videos.

Table 1
The basic information of the testing videos.

	Number of frames	Video duration (s)	Video resolution	Construction machines	Video brightness
Video1	286	11	426×240	1 wheel loader	73.94
Video2	448	14	1280×720	1 excavator, 1 compactor, and 2 dump trucks	53.34
Video3	418	13	640×480	1 excavator, and 1wheel loader	26.43
Video4	450	14	640×480	1 excavator, and 1wheel loader	51.62
Video5	210	30	320×240	1 excavator	32.09
Video6	300	10	1920×1080	2 compactors	44.92
Video7	360	12	1080×608	1 excavator, and 1 dump truck	52.58
Video8	240	10	1280×720	2 compactors	79.65
Video9	280	12	1280×720	2 compactors	90.88

is similar to the proposed method. Meanwhile, this study also tested the proposed method without implementing the illumination enhancement module in order to analyze how the illumination enhancement module can influence the tracking performance in nighttime scenarios. All the three methods (i.e., the proposed method, the proposed method without illumination enhancement, and SORT) were implemented with the same machine detection backbone, this is, the YOLO-v3 algorithm trained on the ACID dataset.

4.3. Evaluation metrics

Six evaluation metrics from the Multiple Object Tracking Benchmark 2016 [59] have been employed in this research: precision, recall, multiple object tracking accuracy (MOTA), multiple object tracking precision (MOTP), the number of ground truth trajectories (GT), and the number of most tracked trajectory (MT). The higher value on the metrics of precision, recall, MOTA, and MOTP indicates the better tracking performance. Of them, precision, recall, and MOTA are generally used to qualify the tracking robustness. The metrics of precision and recall [60] are defined in Eq. (10) and Eq. (11), respectively, which can evaluate the assignment between the tracking outputs and ground truth annotations.

$$Precision = \frac{TP}{TP + FP} \tag{10}$$

$$Recall = \frac{TP}{TP + FN} \tag{11}$$

where, TP (true positive) indicates the number of corrected tracked machines; FP (false positive) means the number of non-machine objects tracked as machine objects; FN (false negative) denotes the number of missing machine objects.

The MOTA (defined in Eq. (12)) combines three sources of tracking errors including false negative, false positive, and the number of tracking ID switch times (IDSW [61]) in the whole trajectory lifespan.

$$MOTA = 1 - \frac{\sum_i (FN_i + FP_i + IDSW_i)}{\sum_i gt_i} \tag{12}$$

where, i is the frame index; gt_i is the number of ground truth machine objects at the frame i; $IDSW_i$ is the number of ID switch times at the frame i; FN_i *and* FP_i have the similar meanings as Eq. (11).

The MOTP measures the tracking precision as Eq. (13), which describes the position similarity between ground truth machine objects and the tracked machine objects.

$$MOTP = \frac{\sum_{i,j} IoU_{i,j}}{\sum_i c_i} \tag{13}$$

where, $IoU_{i,\,j}$ denotes the IoU of machine object j and its corresponding ground truth annotation box at the frame i, and c_i is the number of successfully matched machine objects at the frame i. The larger value of

MOTP indicates the tracker produces more precise tracking windows to localize the machine objects. It is noted that the MOTP is to evaluate the tracking localization ability so that it is different from the precision as introduced in Eq. (10).

GT and MT are used to describe the tracking quality in the object tracking tasks. GT reflects the number of ground truth trajectories of construction machines in a video, and MT shows the number of most tracked trajectories produced by the tracking methods. A trajectory is considered as a MT when the object is correctly tracked over 80% of its lifespan in the video.

4.4. Experimental results

The experimental results in terms of tracking performance using the three different methods are summarized in Table 2. To better demonstrate the experimental process, example images of illumination enhancement results for each testing video have been presented in Fig. 7. Furthermore, Fig. 8 shows example tracking results of Video 4, Video 5, and Video 9 produced by the proposed method. After averaging the tracking performances of nine testing videos, the proposed method achieved an overall precision of 98.80%, recall of 96.30%, the MOTA of 95.10%, and MOTP of 75.90%. Meanwhile, the proposed method successfully tracked 17 of 18 ground truth construction machine trajectories in this experiment. From these tracking results, it can be seen that the proposed method can robustly and precisely track construction machines from nighttime videos. Specifically, it achieved 20.71% higher on recall and 21.70% higher on MOTA than SORT. Also, SORT only tracked 12 machine trajectories, which was less than that of the proposed method. To be noted, the proposed method has achieved higher robustness (MOTA of 95.10%) than exiting tracking methods [27,36] in construction. The MOTA can be further improved by: 1) replacing the current CNN feature extractor ResNet50 with more robust extractor (e.g., ResNet101) to reduce the tracking errors, and 2) increasing the number of training data for backbone detector. By training with more data, the backbone detector will be able to produce more robust results and the MOTA can then be improved.

When removing the illumination enhancement module, the proposed method achieved an overall precision of 98.53%, recall of 79.76%, MOTA of 78.26%, and MOTP of 76.88%. It also successfully tracked 13 construction machine trajectories in the testing videos. Compared with the baseline method, the value of recall and MOTA arrived at 4.17% and 4.86% higher, respectively, which, to some extent, can show the robustness of the proposed method even without implementing the illumination enhancement module. These comparison results also proved that both the illumination enhancement module and the CNN feature similarity can improve the tracking performance when tracking construction machines from nighttime videos. Furthermore, we can see these three tracking methods achieved the similar results (around 75%) on MOTP in this research, which means there is no significant difference in their performance on tracking precision. This is reasonable because the MOTP is determined by the performance of backbone detectors, and all the three methods adopted the same detection algorithm (YOLO-v3) and the training dataset (ACID).

In the experiments, Video 1 and Video 5 only contain the movements of one construction machine, while other videos contain multiple machines. The proposed method has achieved the average MOTA of 93.60% and MOTP of 83.15% on two single construction machine videos, while the average MOTA is 94.77% and MOTP is 78.03% for tracking multiple construction machines. It is found the tracking precision decreased and the tracking robustness slightly increased when the videos had multiple construction machines. Occlusions are considered as another factor that could affect the tracking precision of the proposed method. Video 1, Video 5, Video 8, and Video 9 have no machine occlusions, while other videos contain some degrees of occlusions between construction machines. The proposed method achieved the average MOTP of 86.75% on the four non-occlusion videos, and the average MOTP of 73.10% on the rest of occlusion videos. In other words, the tracking precision of these nighttime videos decreased by 13.65% due to occlusions in this research.

The tracking speed of the proposed method was around 2.1 frames

Table 2
Experimental results of the testing videos.

	Tracking method	Recall	Precision	MOTA	MOTP	GT	MT
Vidoe1	Proposed method	92.00%	100%	92.00%	82.00%	1	1
	Proposed method without IE	88.80%	100%	88.80%	81.40%	1	1
	SORT	85.70%	100%	85.70%	81.20%	1	1
Vidoe2	Proposed method	98.00%	99.20%	97.20%	72.80%	4	4
	Proposed method without IE	94.90%	99.50%	94.20%	72.60%	4	4
	SORT	92.40%	99.50%	81.80%	72.50%	4	3
Vidoe3	Proposed method	99.30%	99.20%	98.50%	72.90%	2	2
	Proposed method without IE	76.80%	98.90%	75.80%	68.30%	2	1
	SORT	76.60%	98.90%	75.60%	68.40%	2	1
Vidoe4	Proposed method	96.40%	97.50%	93.90%	71.10%	2	2
	Proposed method without IE	91.40%	96.80%	88.10%	70.70%	2	2
	SORT	90.90%	96.80%	87.60%	70.60%	2	2
Vidoe5	Proposed method	95.20%	100%	95.20%	84.30%	1	1
	Proposed method without IE	14.80%	100%	14.80%	79.60%	1	0
	SORT	12%	100%	12%	80.90%	1	0
Vidoe6	Proposed method	91.50%	99.80%	91.10%	76.40%	2	2
	Proposed method without IE	69.10%	100%	68.90%	76.30%	2	0
	SORT	57.40%	100%	57%	82%	2	1
Vidoe7	Proposed method	96.70%	96.50%	93.20%	72.30%	2	2
	Proposed method without IE	91.10%	92.80%	84%	70.10%	2	2
	SORT	91.80%	96.80%	88.70%	72.20%	2	2
Vidoe8	Proposed method	100%	99.20%	99.20%	97.40%	2	2
	Proposed method without IE	100%	99.20%	99.20%	87.70%	2	2
	SORT	91.80%	99.10%	90.90%	87.30%	2	2
Video9	Proposed method	90.70%	99.60%	90.30%	83.30%	2	1
	Proposed method without IE	90.90%	99.60%	90.50%	85.20%	2	1
	SORT	81.70%	99.50%	81.30%	85.00%	2	1
Overall	Proposed method	96.30%	98.80%	95.10%	75.90%	18	17
	Proposed method without IE	79.76%	98.53%	78.26%	76.88%	18	13
	SORT	75.59%	98.96%	73.40%	77.79%	18	13

IE refers to illumination enhancement.

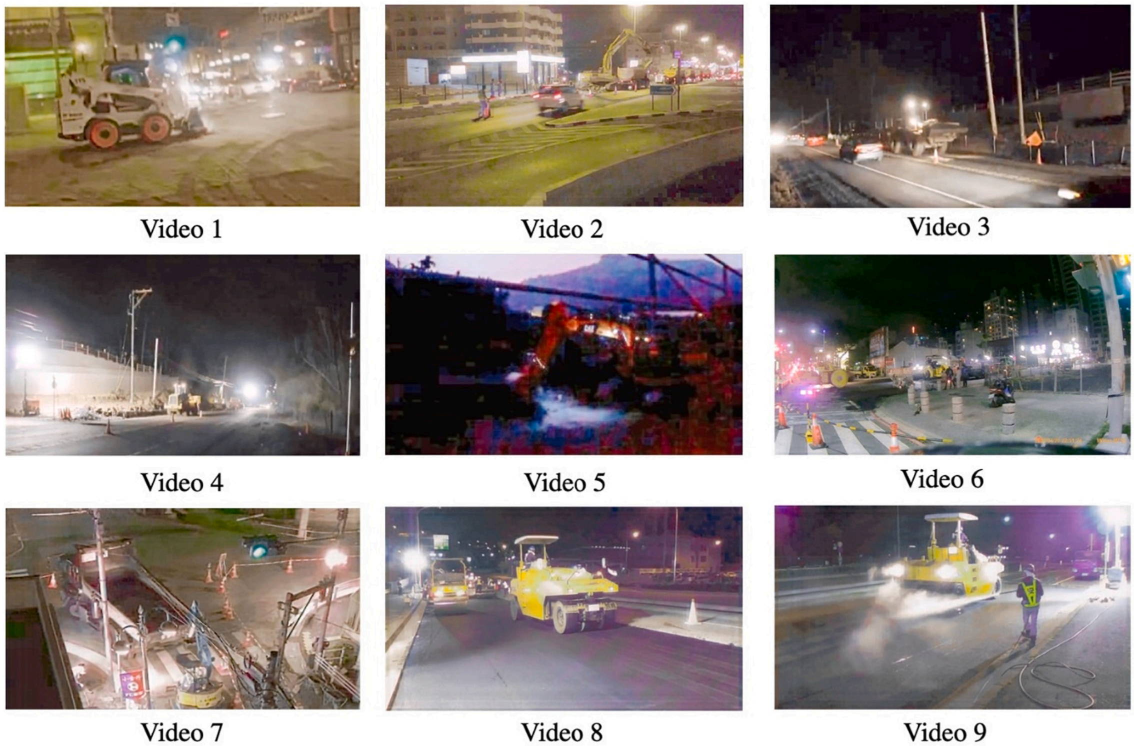


Fig. 7. Example images of illumination enhancement results for testing videos.

Fig. 8. Example images of tracking results produced by the proposed method.

per second (fps) in the current implementation. Due to the processing speed, the proposed method is not suitable to be adopted for real-time tracking applications considering real-time video streaming speed is around 15–30 fps. In such a case, some suggestions could be considered to improve the tracking speed, such as changing the backbone deep learning detection (e.g., changing YOLO-v3 to Tiny-YOLO), adopting better GPU, optimizing the architecture of GLADNet, etc. For practical applications, the tracking speed can also be accelerated by adopting the parallel coding strategy.

5. Discussion

In this section, the experimental results have been analyzed in depth. The discussion includes the influences of the illumination enhancement module and CNN feature extractor. Moreover, the proposed method has been tested on daytime videos to validate its feasibility. The influences of training images dataset have also been discussed.

5.1. Influences of the illumination enhancement module

When tracking construction machines in nighttime videos, the illumination enhancement module becomes more effective as the lighting conditions get worse. To better assess the effectiveness of this module, nine testing videos were first divided into three categories to distinguish different lighting conditions according to the values of video brightness: normal lighting (70–100), insufficient lighting (45–70), and extremely insufficient lighting (0–45). Then, the average tracking performance of the proposed method and the proposed method without illumination enhancement were compared under the above three lighting conditions. The detailed results have been summarized in Table 3. It is obvious to find that the illumination enhancement module could significantly improve the tracking performance under the extremely insufficient lighting conditions. Specifically, the proposed method has achieved 41.76% higher on recall and 41.77% on MOTA than that without implementing the illumination enhancement module. Furthermore, the illumination enhancement module improved 4.57% on recall and 6% on MOTA under insufficient lighting conditions, while the improvement on tracking robustness is only 1% under normal lighting conditions.

When the lighting conditions are extremely insufficient in nighttime videos, the construction machines are difficult to be identified by the deep learning detector and even by human eyes (e.g., images in Fig. 8 (b)). Therefore, the tracking performance is relatively low in extremely insufficient lighting conditions. For example, the proposed method without implementing the illumination enhancement module only achieved 14.80% of MOTP on Video 5, which means the machines cannot be detected in most of the frames. By contrast, after implementing this module, most of frames became detectable and the tracking performance was subsequently improved to a large degree. When the lighting conditions were normal, the deep learning detector can identify construction machines from videos, but the effectiveness of the illumination enhancement module decreased.

5.2. Influences of the CNN feature extractor

Similarly, we also analyzed the effectiveness of the CNN feature extractor by comparing the proposed method without implementing the illumination enhancement module and the SORT method under different lighting conditions. The comparison results can be seen in Table 4. In fact, the major difference between the above two methods is the CNN feature extractor. The proposed method without illumination enhancement achieved 4.97% higher on MOTA than SORT under the extremely insufficient lighting conditions, while the MOTA improvements were 9.34% and 6.96% under the insufficient and normal lighting conditions, respectively. These results can indicate that the CNN feature extractor has stably facilitated the tracking performance in nighttime construction machine tracking, and its effectiveness was not significantly changed under different lighting conditions.

5.3. Influences of daytime testing videos

To better understand the effectiveness of the proposed method, a supplemental experiment has been conducted to test the proposed method on four daytime videos from Xiao and Kang's study [27]. Table 5 summarizes the basic information of these videos and Fig. 9 shows example images of each video. As illustrated in Table 5, the brightness of daytime videos is between 100 and 150. Meanwhile, the brightness of nighttime videos tested in this research is between 20 and 100. This numeric evidence shows that the nighttime videos have different characteristics from daytime videos in terms of brightness.

Table 6 shows the experimental results of the proposed method and SORT on daytime testing videos, while the source testing results of SORT are obtained from [27]. As illustrated in Table 6, the proposed method has achieved 93.00% and 85.98% on MOTA and MOTP, respectively. Compared with the state-of-the-art method SORT, the proposed method shows advantages in terms of tracking robustness (3.08% higher on recall and 3.55% higher on MOTA) on these four testing videos. The experimental results indicate the proposed method can achieve reliable tracking performance on daytime construction videos.

Moreover, the proposed method is considered to be suitable for tracking construction machines from nighttime videos. In this research, the proposed method has achieved the MOTA of 95.10%, MOTP of 75.90%, recall of 96.30%, and precision of 98.80% on nighttime videos. Excepting the metric of MOTP, the proposed method has achieved better performance on metrics of MOTA, recall, and precision on nighttime videos than daytime videos. By contrast, the performance of SORT on nighttime videos has decreased by 16.96% on recall, 16.05% on MOTA, and 7.51% on MOTP compared with those results on daytime videos. The proposed method performs more robust when dealing with nighttime videos because of the deep learning illumination enhancement module, while the SORT cannot maintain its robustness on nighttime tracking.

5.4. Influences of training image datasets

The performance of vision-based tracking methods is related to the image dataset used for training the backbone detector. This research has adopted the ACID dataset for training the YOLO-v3 detector. It is found that the average brightness of 10,000 images in ACID is 125.52. Specifically, 1181 images have a brightness between 0 and 100, 7494 images have a brightness between 100 and 150, and 1325 images have a

Table 3
Influences of the illumination enhancement module on tracking performance under different lighting conditions.

Lighting Condition	Video	Tracking method	Recall	Precision	MOTA	MOTP	GT	MT
Normal lighting	Video 1, Video 9, Video 10	Proposed method	94.23%	99.60%	93.83%	87.57%	5	4
		Proposed method without IE	93.23%	99.60%	92.83%	84.77%	5	4
Insufficient lighting	Video 2, Video 4, Video 7	Proposed method	97.03%	97.73%	94.77%	72.07%	8	8
		Proposed method without IE	92.47%	96.37%	88.77%	71.13%	8	8
Extremely insufficient lighting	Video 3, Video 5, Video 6	Proposed method	95.33%	99.67%	94.94%	77.87%	5	5
		Proposed method without IE	53.57%	99.63%	53.17%	74.73%	5	1

IE refers to illumination enhancement.

Table 4
Influences of the CNN feature extractor on tracking performance under different lighting conditions.

Lighting Conditions	Videos	Tracking method	Recall	Precision	MOTA	MOTP	GT	MT
Normal lighting	Video 1, Video 9, Video 10	Proposed method without IE	93.23%	99.60%	92.83%	84.77%	5	4
		SORT	86.40%	99.53%	85.97%	84.50%	5	4
Insufficient lighting	Video 2, Video 4, Video 7	Proposed method without IE	92.47%	96.37%	88.77%	71.13%	8	8
		SORT	84.13%	98.77%	79.43%	73.13%	8	5
Extremely insufficient lighting	Video 3, Video 5, Video 6	Proposed method without IE	53.57%	99.63%	53.17%	74.73%	5	1
		SORT	48.67%	99.63%	48.20%	77.10%	5	2

IE refers to illumination enhancement.

Table 5
The basic information of four daytime videos (adopted from Xiao and Kang's study [27]).

	Number of frames	Video duration (s)	Video resolution	Construction machines	Video brightness
Daytime_Video1	1250	50	1280×720	1 dump truck, and 1 wheel loader	136.11
Daytime_Video2	1050	35	1280×720	1 dump truck, and 1 wheel loader	103.74
Daytime_Video3	1200	48	1280×720	5 dump trucks, and 2 excavators	114.81
Daytime_Video4	1000	40	1280×720	5 dump trucks, and 3 excavators	129.19

Fig. 9. Example images of four daytime videos (adopted from Xiao and Kang's study [27]).

Table 6
Experimental results of the daytime testing videos.

	Tracking method	Recall	Precision	MOTA	MOTP	GT	MT
Vidoe1	Proposed method	99.50%	99.90%	99.40%	87.10%	2	2
	SORT	100.00%	98.40%	98.40%	87.30%	2	2
Vidoe2	Proposed method	91.70%	91.70%	83.40%	87.50%	2	2
	SORT	87.90%	91.20%	79.30%	86.00%	2	1
Vidoe3	Proposed method	95.30%	98.50%	93.60%	83.80%	7	6
	SORT	89.50%	97.70%	88.00%	82.10%	7	6
Vidoe4	Proposed method	96.00%	99.60%	95.60%	85.50%	9	8
	SORT	92.80%	98.40%	92.10%	85.80%	9	8
Overall	Proposed method	95.63%	97.43%	93.00%	85.98%	20	18
	SORT	92.55%	96.43%	89.45%	85.30%	20	17

brightness between 150 and 255 in ACID. The images captured at nighttime usually have the brightness smaller than 100. Considering most images in ACID are captured in daytime, the brightness and lighting conditions in ACID are different from those in nighttime videos.

Currently, there did not exist studies that tracked construction resources at nighttime with training the backbone detector by image datasets collected at nighttime, while most construction image datasets [55,62,63] are focusing on daytime scenarios. When adopting existing construction image datasets, the tracking methods are naturally more suitable for daytime tracking rather than nighttime tracking. For example, the SORT method achieves better performance on robustness (16.05% higher on MOTA) and precision (7.51% higher on MOTP) on daytime videos. Developing a nighttime construction image dataset is a possible way of achieving reliable nighttime tracking. By adopting the nighttime image dataset, the backbone detector is expected to produce more stable detection results from nighttime videos and the tracking performance will be eventually improved. However, building such a dataset is time-consuming and costly. To avoid the efforts required for developing a new dataset, this research solves the nighttime construction machine tracking by introducing the deep learning illumination enhancement module. As such, the proposed method has achieved robust performance on nighttime tracking by adopting existing well-developed construction image datasets.

The proposed method has achieved the MOTP of 75.90%, which is about 5% to 15% lower than the performance of state-of-the-art vision-based tracking methods on daytime scenarios [27,36]. This is mainly because the brightness conditions of the existing training datasets are different from the nighttime videos. As such, the backbone detector is difficult to produce precise detection results from the nighttime videos. Also, the feasibility of the backbone detector affects the tracking precision. To be noted, the MOTP of 75.90% can still satisfy the purpose of construction machine tracking. As reported in [59], five testing tracking methods have achieved the MOTP of 76.04% on the MOT16 benchmark, while their performance is considered to be satisfactory for object tracking. The tracking precision of the proposed method can be improved from two perspectives including: 1) replacing YOLO-v3 with a more precise deep learning detector, such as Faster RCNN [64], and 2) collecting more nighttime construction images and adding them to the training dataset. In this way, the backbone detector will be able to produce more precise detection results for tracking.

6. Conclusions and future works

This research presents a vision-based method for automatic tracking of construction machines in nighttime videos. Specifically, the proposed method adopts the deep learning illumination enhancement method to improve the lighting conditions in nighttime videos and a CNN feature extractor as the appearance model to distinguish construction machines. After applying this approach to nine nighttime testing videos, the tracking performance of construction machines was relatively satisfactory. For instance, the average MOTA and MOTP arrived at 95.10% and 75.90%, respectively. Meanwhile, 17 of 18 construction machine trajectories were successfully tracked by the proposed method. Under extremely insufficient lighting conditions, the illumination enhancement module sill can improve the tracking robustness around 41% in recall and MOTA. By adopting the proposed method, construction machines can be robustly tracked in nighttime videos and several construction applications can be further automated in nighttime scenarios, such as the crew productivity analysis and struck-by accidents monitoring.

This research contributes to the body of knowledge of vision-based construction resources monitoring from three perspectives. Firstly, this research provides a robust method for tracking construction machines at nighttime. By adopting the proposed method, a large number of automated monitoring applications (e.g., machine idling monitoring and machine collision alarm) can be directly applied to nighttime construction scenarios without requiring additional efforts. Secondly, to the authors' knowledge, this is the first research that integrated the deep learning illumination enhancement method to vision-based construction resources tracking. The experimental results show the potential of illumination enhancement methods in nighttime construction sites monitoring. Thirdly, this research proves the feasibility of the CNN feature extractor as the appearance model in nighttime tracking of construction machines.

The future work will focus on improving the tracking speed and precision of the proposed method and more backbone deep learning detectors will be tested for this purpose. Currently, this research is only suitable for tracking construction machines from nighttime videos. More works will be conducted on tracking construction workers in order to avoid the potential collisions between workers and construction machines. Another future work is to combine the proposed method with sensor-based methods (e.g., the global positioning system) to improve its performance of dealing with occlusions. Finally, more efforts will be conducted to develop construction image datasets that were collected at nighttime.

Declaration of Competing Interest

No potential conflict of interest was reported by the authors.

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